

STEGANOGRAPHY

INTRODUCTION

Steganography is the art and science of hiding information within other non-secret data or media to avoid detection. Unlike encryption, where the message is encoded and visible, steganography conceals the very existence of the message. In a lab project, steganography often focuses on embedding secret data into digital media like images, audio, or video files.

REQUIREMENT

1. The application accept an image file say along with the a text which contains the message to be steged.
2. Analyze the size of the message file and the data part of the file to check whether the message could fit in the provided image.
3. Provide a option to steg a magic string which could be usefull to identify whether the image is steged or not.
4. The application should provide a option to decrypt the file.
5. This has to be an command line application and all the options has to be passed as an command line argument.

USES

1. Secure Communication: Sending confidential data without revealing its existence.
2. Digital Watermarking: Protecting intellectual property by embedding copyright information in media.
3. Forensic Analysis: Embedding tracking data for investigative purposes.
4. Data Authentication: Validating the authenticity and integrity of digital media.

NEED

1. **Privacy:** Ensures sensitive information is shared discreetly.
2. **Data Integrity:** Protects information from tampering during transmission.
3. **Efficient Storage:** Embeds data without significantly increasing file size.
4. **Stealth:** Minimizes the likelihood of detection by maintaining media quality.